

Casey Stella

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PROFESSIONAL **Netflix**, Los Gatos, CA

EXPERIENCE

ML Engineer

March 2024 – Present

Work on Netflix's core recommendation systems, with a focus on row ranking and the models and algorithms that determine how recommendations are organized and presented. Contribute directly to the core ranking algorithms, including improving model architectures, training objectives, and model internals. Help evolve the existing ranking approach toward a large generative recommendation model, adapting established recommendation techniques to a substantially different modeling architecture and improving its ranking quality and behavior.

Stripe, San Francisco, CA

Staff Software Engineer

March 2019 – March 2024

Tech lead for Stripe's Machine Learning Infrastructure organization, responsible for the platform used to train, deploy, and serve hundreds of production ML models under tight latency and availability SLAs. Designed and operated infrastructure for scalable model serving, training, feature generation, and feature engineering. Advocated for and helped establish Stripe's LLM infrastructure effort, building internal LLM serving infrastructure and applying large language models to production problems, including merchant classification and detection of businesses selling prohibited or restricted products as well as a large scale text to SQL internal product. Note: I personally worked on these, other use-cases with LLMs were in flight.

Finite State, Columbus, OH

Vice President of Data

August 2018 – January 2019

As Vice President of Data at Finite State, a Series A stage cybersecurity startup, I was in charge of scalable data architecture and machine learning. I operated as both an individual contributor as well as architectural support.

Apache Software Foundation,

Vice President of Apache Metron

April 2017 – November 2020

In my role as a committer and contributor to Apache Metron, funded by Hortonworks, I also performed the role of Vice President of the Apache Metron project for the Apache Software Foundation. During this role, I shepherded the project from incubation through to being a top level project. During this period, I managed releases, spoke about the project at public conferences and contributed heavily to its development and advancement.

A small selection of talks that cover my work on Apache Metron:

- Apache Metron as a Case Study of a Modern Streaming Architecture on Hadoop
- Model as a Service: Modern Streaming Data Science with Apache Metron

Hortonworks, San Jose, CA

Principal Software Engineer

December 2012 – July 2018

As a principal engineer and data scientist on the Apache Metron team, I lead the advanced analytics team. I was tasked with developing the next generation data science and machine learning infrastructure in Apache Metron. Apache Metron is an open source, scalable cybersecurity analytics platform that is maintained within the Apache Software Foundation.

Principal Architect

November 2012 – December 2015

As a Principal Architect in the consulting services organization at Hortonworks, I provide mentorship and guidance in operationalizing Hadoop and its ecosystem to solve business problems for our clients. I work with our clients to reduce risk and time to market by providing expertise in big data.

- Provide proof-of-concepts to reduce engineering churn
- Give extensive presentations about the Hadoop ecosystem, best practices, data architecture in Hadoop
- Provide mentorship and guidance to other architects to help them become independent
- Provide review and feedback for existing physical architecture, data architecture and individual code
- Debug and solve issues with Hadoop as on-the-ground subject matter expert. This could include everything from patching components to post-mortem analysis of errors.

I focus on issues around data science and especially natural language processing at scale. I have domain knowledge in medical/clinical informatics and oil/gas data analysis and signal processing.

Explorys, Cleveland, OH

“Big Data” Architect

January 2011 – November 2012

I was a “big data” architect and, prior to that, a senior software engineer on the platform team at Explorys. The team was responsible for the creation, care and maintenance of the high performance indexing infrastructure. My job required a deep understanding of the Hadoop ecosystem. I designed the next generation data architecture for the unstructured data at Explorys as well as writing, debugging, and analyzing the performance of many map reduce jobs to realize that architecture.

- Devised and lead the implementation of the next generation architecture for more efficient data ingestion and processing.
- Proficiency with mentoring and on-boarding new engineers who are not proficient in Hadoop and getting them up to speed quickly.
- Experience with being a technical lead of a team of engineers.
- Proficiency with modern natural language processing and general machine learning techniques and approaches
- Extensive experience with Hadoop and HBase, including multiple public presentations about these technologies.
- Experience with hands on data analysis and performing under pressure.
- Designed and wrote a layer on top of MapReduce to make the task of writing MapReduce jobs easier and more safe for Junior Engineers.

Game Communications, Mayfield Heights, OH

Senior Engineer

November 2009 – January 2011

Designed, implemented and integrated into legacy code a scalable network infrastructure layer for an instant message and VOIP network

- Created a software routing layer and implementing an ordered, reliable protocol on top of the raw network layer
- Integrated Cassandra as a distributed persistent metadata store to provide metadata resolution for network entities on the network
- Created low-level network libraries in C++ and C# to send messages across network
- Spearheaded institution of pair-programming and semi-agile practices

ION Geophysical, Houston, TX

Research Geophysicist

October 2008 – November 2009

Member of the time processing R&D team, tasked to design, prototype and implement production-ready algorithms to use applied math and signal processing to reduce noise in seismic data. This was a telecommuting position from Cleveland, OH.

- Implemented multithreaded C++ applied linear algebra library consisting of efficient STL-compliant data structures and algorithms to assist in signal processing
- Assisted in the design and implementation of an internal flow-based map-reduce system for doing distributed scientific computation
- Used Tokyo Cabinet to provide an on-disk index of seismic metadata in an efficient way
- Implemented a multidimensional least squares adaptive filter that decreased wall-clock time for our most common seismic task by an order of magnitude
- Developed a fixed-precision algorithm in C for the embedded PowerPC 405 platform which uses first-order statistics to differentiate noise from initial signal in an accurate, efficient and robust way
- Spearheaded institution of pair-programming and semi-agile practices in a geographically dispersed environment

Oracle, Cleveland, OH

Senior Member of Technical Staff

October 2005 – October 2008

Member of the Oracle Enterprise Repository team. We provided an enterprise J2EE application to organize and manage assets and their metadata.

- Worked as part of a geographically distributed team using agile practices
- Designed and implemented an enterprise build system around Maven, porting an existing heterogeneous build system from Ant and shell scripts to Maven
- Implemented multiple performance and scalability improvements, resulting in substantial benefits deriving from better caching strategies and algorithms with more favorable CPU/Memory complexity characteristics.
- Managed summer interns

CERTIFICATION	Cloudera Certified Hadoop Developer Cloudera Certified HBase Developer	August 2012 August 2012
EDUCATION	Texas A&M University , College Station, TX <i>Masters of Science in Mathematics</i> • Emphasis in Computational Complexity and Theoretical Computer Science • Advised by Dr. J. Maurice Rojas University of Louisiana at Monroe , Monroe, LA <i>Bachelors of Science in Mathematics and Computer Science</i>	Spring 2005 Spring 2002
SELECTED PUBLICATIONS	Ofer Mendelevitch, Douglas Eadline, and Casey Stella “Practical Data Science with Hadoop and Spark: Designing and Building Effective Analytics at Scale“, Book by <i>Addison-Wesley</i> , 2016. J. Maurice Rojas, Frederic Bihan, and Casey Stella “Faster Real Feasibility via Circuit Discriminants“, In <i>Proceedings of ISSAC 2009</i> , pp. 39-46, ACM Press, 2009.	
PATENTS	“Targeted Optical Character Recognition (OCR) for Medical Terminology“, <i>Patent Issued June 7, 2016</i> , Patent number 9361531, Optum Technologies “SYSTEM AND METHOD FOR USING AN EDITABLE LIFECYCLE EVENT DISTRIBUTION LIST WITH A SERVICE METADATA REPOSITORY“, <i>Patent Issued March 27, 2012</i> , Patent number 8145680, Oracle Corporation	
SELECTED TALKS	I was a regular at the Hadoop Summit, Spark Summit EU and QCon London. You can find the slides for my various talks here. “New Complexity Thresholds for Sparse Real Polynomials“, <i>Sixth International Joint Meeting of AMS and SMM</i> , Invited Talk	
HONORS AND AWARDS	Outstanding Teaching Assistant [2005] National Science Foundation VIGRE Fellowship [2004] AUF & Regents Fellowships for Outstanding Academic Achievement [2003]	